

(idealni)
 5.8 Plin prevodniških e^- v Ag

$$T=0$$

$$\rho = 10500 \text{ kg/m}^3$$

1 prevod. e^- /atom

$$M = 107,9 \text{ kg/kmol}$$

$$\mu = ?$$

$$p = ?$$

$$x_T = ?$$

$$E_j = \frac{p_j^2}{2m} \rightarrow p_j = \sqrt{2E_j m} \quad p_j dp_j = m dE_j$$

$$N = \sum_i \frac{1}{1 + \exp(\beta(E_j - \mu))} = \sum_j f(E_j)$$

$$\Sigma \rightarrow \int$$

$$\begin{aligned} \text{spin} \rightarrow 2 \frac{\int d\Gamma}{h^3} &= 2 \frac{\int 4\pi p^2 dp dV}{h^3} = \frac{V 8\pi}{h^3} \int \sqrt{2mE} m dE = \\ &= \underbrace{\frac{V \sqrt{2} m^{3/2}}{h^3 \pi^2}}_{g(E)} \int \sqrt{E} dE \end{aligned}$$

$$\rightarrow N = \int_0^\infty f(E) g(E) dE \underset{T=0}{=} \int_0^\mu g(E) dE = \frac{\sqrt{2} V m^{3/2}}{h^3 \pi^2} \int_0^\mu E^{1/2} dE = \frac{2\sqrt{2}}{3} \frac{V m^{3/2}}{h^3 \pi^2} \mu^{3/2}$$

$$\mu = \frac{h^2}{2m_e} \left(\frac{3N\pi^2}{V} \right)^{2/3} = \underline{\underline{5,5 \text{ eV}}}$$

$$\frac{N}{V} = \frac{\rho N_a}{M}$$

$$-\beta q = \beta p V = \sum_j \ln(1 + \exp(-\beta(E_j - \mu))) = \frac{\sqrt{2} V m^{3/2}}{h^3 \pi^2} \int_0^\infty \sqrt{E} \underbrace{\ln(1 + \exp(-\beta(E - \mu)))}_I dE$$

$$I = \frac{2}{3} E^{3/2} \cdot \ln(1 + \exp(-\beta(E - \mu))) \Big|_0^\infty + \frac{2}{3} \int_0^\infty \frac{E^{3/2} \beta dE}{1 + \exp(\beta(E - \mu))}$$

$\downarrow$ LH (zg. meja)

$$\frac{2}{3} \frac{-\beta}{(-\frac{3}{2}) E^{-5/2} (1 + \exp(\beta(E - \mu)))} \Big|_{E \rightarrow \infty} = 0$$

$$du = \frac{-\beta \exp(-\beta(E - \mu))}{1 + \exp(-\beta(E - \mu))} dE = \frac{-\beta dE}{1 + \exp(\beta(E - \mu))}$$

$$V = \frac{2}{3} E^{3/2}$$

$$\rightarrow I = \frac{2}{3} \int_0^{\mu} E^{\frac{3}{2}} \beta dE = \frac{4}{15} \beta \mu^{\frac{5}{2}}$$

$$\beta p V = \beta \cdot \frac{2^{\frac{5}{2}}}{15} \mu^{\frac{5}{2}} \frac{N m_c^{\frac{3}{2}}}{h^3 T^2} = \underline{\underline{2 \cdot 10^{10} \text{ Pa}}}$$

$$\chi_T = -\frac{1}{V} \left(\frac{\partial U}{\partial p} \right)_T$$

$$p \propto \mu^{\frac{5}{2}}, \mu \propto V^{-\frac{2}{3}} \Rightarrow p \propto V^{-\frac{5}{3}} \rightarrow pV^{\frac{5}{3}} = k \rightarrow \ln p + \frac{5}{3} \ln V = \ln k$$

$$\frac{dp}{p} + \frac{5}{3} \frac{dV}{V} = 0$$

$$\chi_T = -\frac{1}{V} \left(-\frac{3}{5} \frac{V}{p} \right) = \frac{3}{5} \frac{1}{p} = \underline{\underline{2.9 \cdot 10^{-11} \text{ Pa}^{-1}}}$$

$$\frac{dV}{dP} = -\frac{3}{5} \frac{V}{P}$$

5.1 Valovanje z disp. relacijo $\omega = ak^n$, nizkot obnašanje c (visoko)

zvoč: $n=1, \omega = ck$

$\mu=0$
fotoni (delci, ki se obn. kot fotoni pri svetlobi)

spinski valovi: $n=2, \omega = ck^2$

magnoni

$$c = \frac{d\langle E \rangle}{dT}$$

$$\langle E \rangle = \sum_j \langle N_j \rangle \epsilon_j = \left(\sum_j \left(\langle N_j \rangle + \frac{1}{2} \right) \hbar \omega_j \right)$$

$$= \sum_j \hbar \omega_j \frac{1}{\exp(\beta \epsilon_j) - 1}$$

za c hima pomembna

polarizacija

$$\sum_j \rightarrow \int \frac{d\Gamma}{h^3} = P \frac{V}{h^3} \int 4\pi p^2 dp = P \frac{4\pi V}{h^3} \int \hbar^3 k^2 dk = \frac{PV}{2\pi^2} \int k^2 dk =$$

$$\left[\begin{aligned} k &= \left(\frac{\omega}{a} \right)^{\frac{1}{n}} \\ dk &= \frac{1}{na} \left(\frac{\omega}{a} \right)^{\frac{1}{n}-1} d\omega \end{aligned} \right] = \frac{PV}{2\pi^2} \int \left(\frac{\omega}{a} \right)^{\frac{2}{n} + \frac{1}{n} - 1} \frac{1}{na} d\omega = \frac{VP}{2\pi^2 n a^{\frac{2}{n} + \frac{1}{n}}} \int \omega^{\frac{1}{n}-1} d\omega$$

dovolj velik vzorec

$$\langle E \rangle = \frac{VP\hbar}{2\pi^2 n a^{\frac{2}{n}}} \int_{\omega_{\min}}^{\omega_{\max}} \frac{\omega^{\frac{1}{n}}}{\exp(\beta \hbar \omega) - 1} d\omega = \frac{VP}{2\pi^2 n a^{\frac{2}{n}}} \frac{1}{(\beta \hbar)^{\frac{1}{n}} \beta} \int_{u_{\min}}^{u_{\max} = \beta \hbar \omega_{\max}} \frac{u^{\frac{1}{n}}}{e^u - 1} du$$

1) niska T: $\beta \rightarrow \infty$, $u_{\max} \rightarrow \infty$

$$I = \int_0^{\infty} \frac{u^{3/2}}{e^u - 1} du = \int_0^{\infty} \frac{e^{-u} u^3}{1 - e^{-u}} du = \int_0^{\infty} e^{-u} u^3 \sum_{h=1}^{\infty} (e^{-u})^h du = \sum_{h=1}^{\infty} \int_0^{\infty} u^3 e^{-hu} du = \sum_{h=1}^{\infty} \frac{1}{h^4} \Gamma(4)$$

zvok ($n=1, a=c, p=3$)

$$= 6 \sum_{h=1}^{\infty} \frac{1}{h^4} = 6 \cdot \zeta(4) = \frac{6}{90} \pi^4 = \underline{\underline{\frac{\pi^4}{15}}}$$

$$\langle E \rangle = \frac{3V}{2\pi^2 c^3} \frac{(k_B T)^4}{h^3} \cdot \frac{\pi^4}{15 \cdot 5}$$

$$C = \frac{k_B^4 V \pi^2}{2h^3 c^3 5} 2k_B T^3 \propto \underline{\underline{T^3}}$$

2) visoka T, zvok:

$$\langle E \rangle = \frac{3V}{2\pi^2} \frac{h}{c^3} \frac{k_B^4 T^4}{h^4} \int_0^{u_{\max}} \frac{u^3 du}{e^u - 1}$$

$$I = \int_0^{u_{\max}} \frac{u^3 du}{1 + u - 1}$$

$$\langle E \rangle = \dots = 3N k_B T$$

$$\underline{\underline{C = 3N k_B}}$$

Debye: $w = ck$

$$3N = 3V \int_0^{p_{\max}} \frac{4\pi p^2 dp}{h^3} = \frac{12\pi V h^3}{h^3} \int_0^{u_{\max}} \frac{u^2}{c^3} du$$

$$\rightarrow w_{\max} = \frac{3N \cdot 2\pi^2 c^3}{V}$$

6.4) Okrogla bučka z vodno paro & ohlajenim delom stene



$$p(t) = n(t) k_B T$$

$$j = n \frac{\langle v \rangle}{4} = -\frac{1}{A} \frac{dN}{dt} = -\frac{V}{A} \frac{dn}{dt}$$

$$\int_{n'}^n \frac{dn}{n} = -\frac{A}{V} \int_0^t \frac{\langle v \rangle}{4} dt$$

$$\ln \frac{n(t)}{n'} = -\frac{A}{V} \frac{\langle v \rangle}{4} t$$

$$n(t) = n' e^{-\frac{A \langle v \rangle}{V} t}$$

$$p = n' k_B T e^{-\frac{A \langle v \rangle}{V} t} = p' e^{-\frac{A \langle v \rangle}{V} t}$$

$$\ln \frac{p}{p'} = \ln \frac{n}{n'} = -\frac{A}{4V} \sqrt{\frac{8 k_B T}{\pi m}} t$$

$$\rightarrow A = -\frac{4V}{t} \sqrt{\frac{\pi m}{8 k_B T N_A}} \ln \frac{p}{p'} = \underline{\underline{4,7 \text{ cm}^2}}$$

$$V = 10 \text{ dm}^3$$

$$T = 20^\circ \text{C} = \text{konst}$$

$$p' = 10 \text{ mbar}$$

$$t = 1 \text{ s}$$

$$p = 10^{-3} p'$$

$$M = 18 \text{ kg/kmol}$$

$$A = ?$$